GCI Final Assignment

# Capital Profit Model : AI-Lifted Telecom Churn & Retention Optimizing

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**Citations and References (Part 1):**
[1] Bhushan01 notebook collab: https://colab.research.google.com/drive/1RCvCkSZhmwsI-0ualzFdHSC7qRzS2VJ2
[2] Final Assignment Folder: https://drive.google.com/drive/folders/1UxmN3o99Ed5r3lwnMVVcT7fi0h4slOvc?usp=drive_link
[3] https://www.intelmarketresearch.com/telecommunications-market-11782
[4] https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-network-is-the-product-how-ai-can-put-telco-customer-experience-in-focus
[5] https://www.transparencymarketresearch.com/telecommunication-market.html

**Citations and References (Part 2):**
[6] https://tridenstechnology.com/telecom-churn/
[7] https://gemini.google.com/share/25085054928c
[8] https://chatgpt.com/share/6a33ff2f-7228-83ee-ae0d-808969e8d4af
[9] https://gemini.google.com/share/df52a0fe5c40
**[10] Image Source: https://gemini.google.com/share/25085054928c**

# EXECUTIVE SUMMARY

## The Problem Statement

### The Retention Dilemma

Our telecom portfolio is suffering from highly predictable, preventable revenue leakage through customer churn.

Our analysis reveals that **hardware obsolescence**—not network quality—is the true catalyst for this defection.

The core business bottleneck is financial: ignoring churn guarantees millions in lost recurring revenue, but offering a blanket $30 device subsidy to the entire customer base to stop it will bankrupt the retention budget.

## What We Are Going To Do

### Surgical AI Intervention

We will transition the company from a reactive, mass-market retention approach to a proactive, data-driven strategy.

By deploying an advanced predictive machine learning engine (**Optimized XGBoost**), we will isolate customers who:

- Are highly likely to leave imminently due to aging devices.
- Represent high ongoing monthly recurring revenue.
- Are highly likely to accept a targeted $30 handset upgrade.

## The Financial Impact

### Maximized Enterprise ROI

We have translated the model's complex predictions into a simple, actionable business framework: the **Customer Risk & Revenue Index (CRRI)**.

By using the CRRI to rank and deploy subsidies *only* to the highest-value, highest-risk accounts, we transform retention into a scalable profit engine.

This targeted campaign is projected to generate an estimated **$2.8M in scaled net enterprise profit** (per 100k users over a 6-month horizon).

Source: Bhushan01, Google Colab Notebook, Cell: "Section 4: ALGORITHMIC VALIDATION & FINANCIAL SIMULATION"
“Section 15: ADVANCED ANALYTICS & CRRI FRAMEWORK”

# MARKET CONTEXT

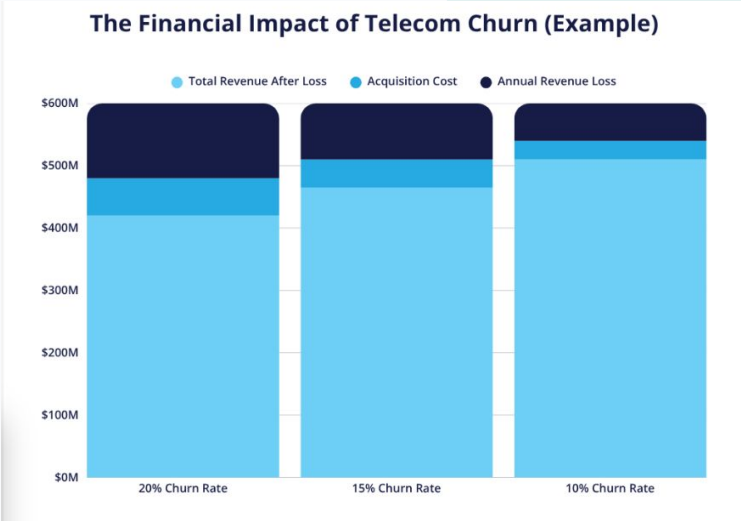
## The Market Situation: Massive Scale but Saturated Growth

- The global telecommunications market reached a valuation between $1.63 Trillion and $1.9 Trillion in 2024.
- Despite this massive scale, market saturation severely limits organic expansion, with the global CAGR projected at a modest 1.6% to 5.3%.
- Profitability must now be driven by protecting the existing subscriber base rather than relying on net-new customer acquisitions.

## Analyzing the Risks: Severe Defection and Exorbitant Costs

- The sector is currently battling high volatility, with average annual churn rates hovering between 20% and 31%.
- Acquiring a new customer costs 5 to 10 times more than retaining an existing one, posing a critical financial risk.
- Reactive, blanket marketing campaigns are financially unsustainable against these high acquisition costs.

Source: [4] https://tridenstechnology.com/telecom-churn/




## Assumptions & AI Needs

- 5G network acceleration creates extreme defection risks.
- Providers must focus on targeted retention incentives.
- Predictive analytics can cut churn by up to 15%.


**Sources:**
[1] https://www.intelmarketresearch.com/telecommunications-market-11782
[2] https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-network-is-the-product-how-ai-can-put-telco-customer-experience-in-focus
[3] https://www.transparencymarketresearch.com/telecommunication-market.html

# The Baseline Analysis – Quantifying the Financial Toll of Inaction

Section 5 of collab notebook explicitly explains the proof and workflow of this numbers !!
Data used was from Client,Record .csv

**Portfolio Baseline**

**49.56%**

Overall Churn Rate

**$57.91**

Calculated ARPU

## The Churn Reality: A Leaking Portfolio

- Historical portfolio baseline suffers from an unoptimized and highly predictable structural churn rate.
- Constant defection drains active customer base, neutralizing expensive new acquisition capital.

## Portfolio Value: The Revenue at Risk

- Median total revenue (totrev) represents a massive, established capital asset left unprotected.
- Mean monthly account value stands at a highly lucrative **$57.91 per month**.

## The Economic Cost of the Status Quo

- Losing one average account costs the enterprise **$347.46 (=6*$57.91)in gross revenue** over 6 months.
- Compounded impact results in a direct multi-million dollar hit to enterprise EBITDA margins.

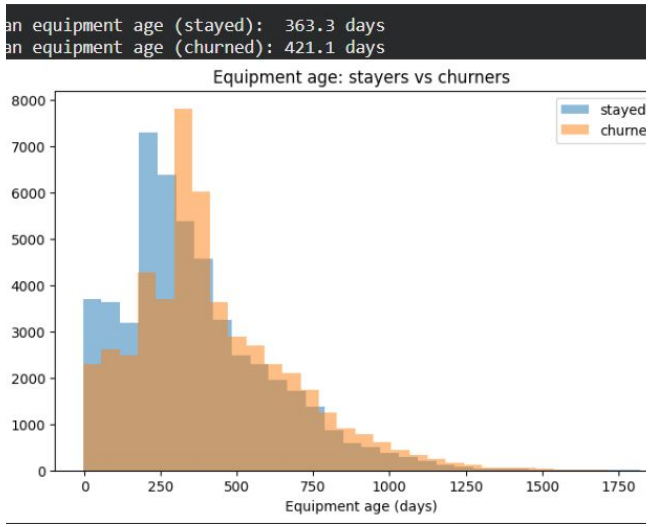
[1] Source: Bhushan01, Google Colab Notebook, Cell: Section 12: Decile Analysis, Section 2: EXPLORATORY DATA ANALYSIS (SLIDE METRIC PROOFS)

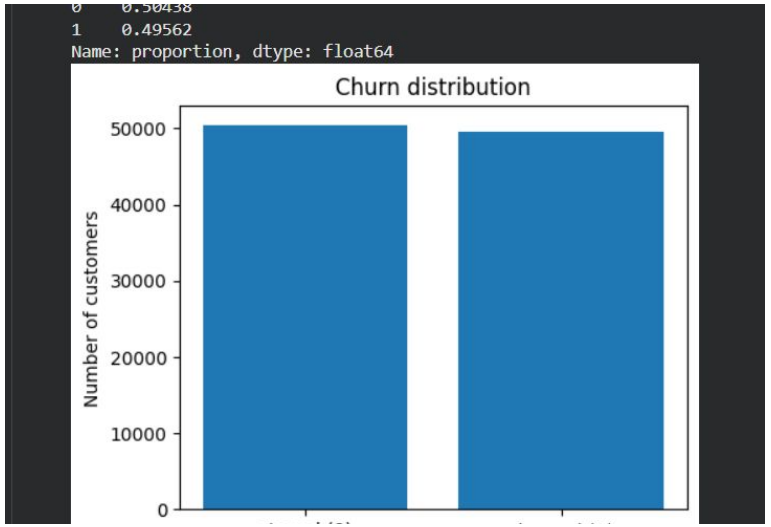
# EDA : DEBUNKING THE MYTH

**The Network Performance Misconception:** Conventional enterprise strategy assumes that technical friction—specifically dropped calls and network instability—is the primary reason customers port out to competitors.

## The Analytical Reality

An aggregation of monthly voice drop behaviors reveals that **retained users** and **churned users** show a statistically negligible delta.



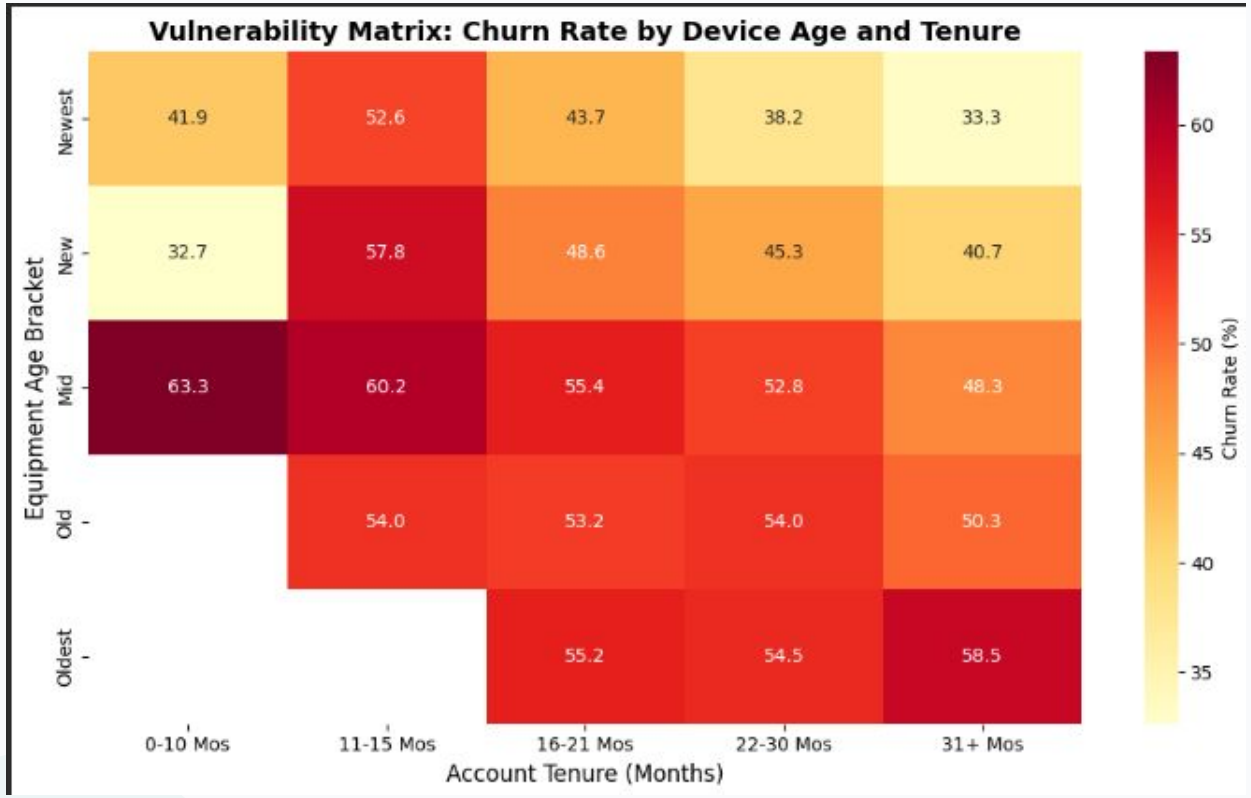

**Strategic Imperative:** Investing capital into massive infrastructure or network optimization projects *specifically to solve churn* will yield a near-zero ROI. Our customers are not leaving due to call drops; we must look elsewhere for the true economic driver.

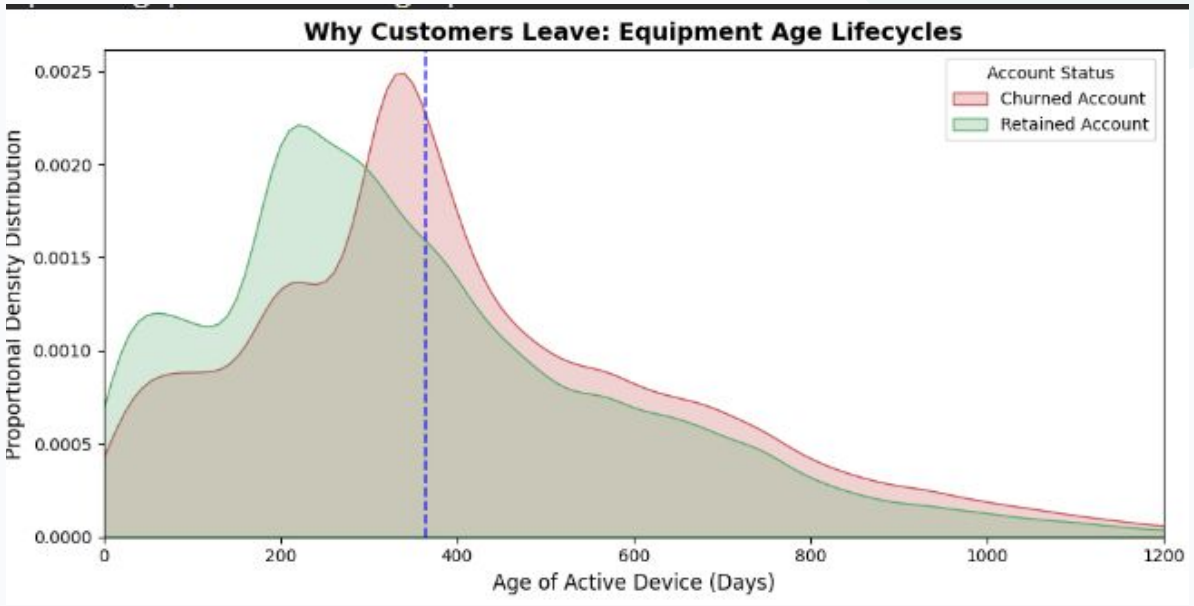
Source: Bhushan01, Google Colab Notebook, Section 2: EXPLORATORY DATA ANALYSIS (SLIDE METRIC PROOFS)

Source: Bhushan01, Google Colab Notebook,Section 5: VISUAL GRAPHIC ASSET PRODUCTION: Section 10: DIAGNOSTIC RISK HEATMAP Section 12: Decile Analysis:

# EDA : THE TRUE CULPRIT

## Isolating Handset Obsolescence and the "12-Month Cliff"




### The Impact of Hardware Age

Network quality is a non-factor; mobile device age (eqpdays)is the definitive vector driving defection. Retained: 406 days avg. vs. Churned: 508 days avg.

### The "12-Month Cliff" Phenomenon

Granular lifecycle segmentation proves there is a severe vulnerability window concentrated precisely between months 11 and 15 of an account's tenure.

### The Behavioral Trigger

At the one-year mark, customers experience natural device degradation (battery wear, processor slowdown) and reach the end of standard contract milestones. Competitors exploit this specific window with aggressive new device advertising. If we fail to intercept them with a hardware upgrade during this critical window, they port out.

# PROBLEM DEFINITION

**The Business Objective:** Retain high-value customers by offering them a proactive hardware upgrade before they defect.

## Strategic Target Variable Selection

### The "Goldilocks Zone"

The target variable `churn` identifies customers leaving 31-60 days after observation.

**0-30 days?**

Too late. Customer likely already porting.

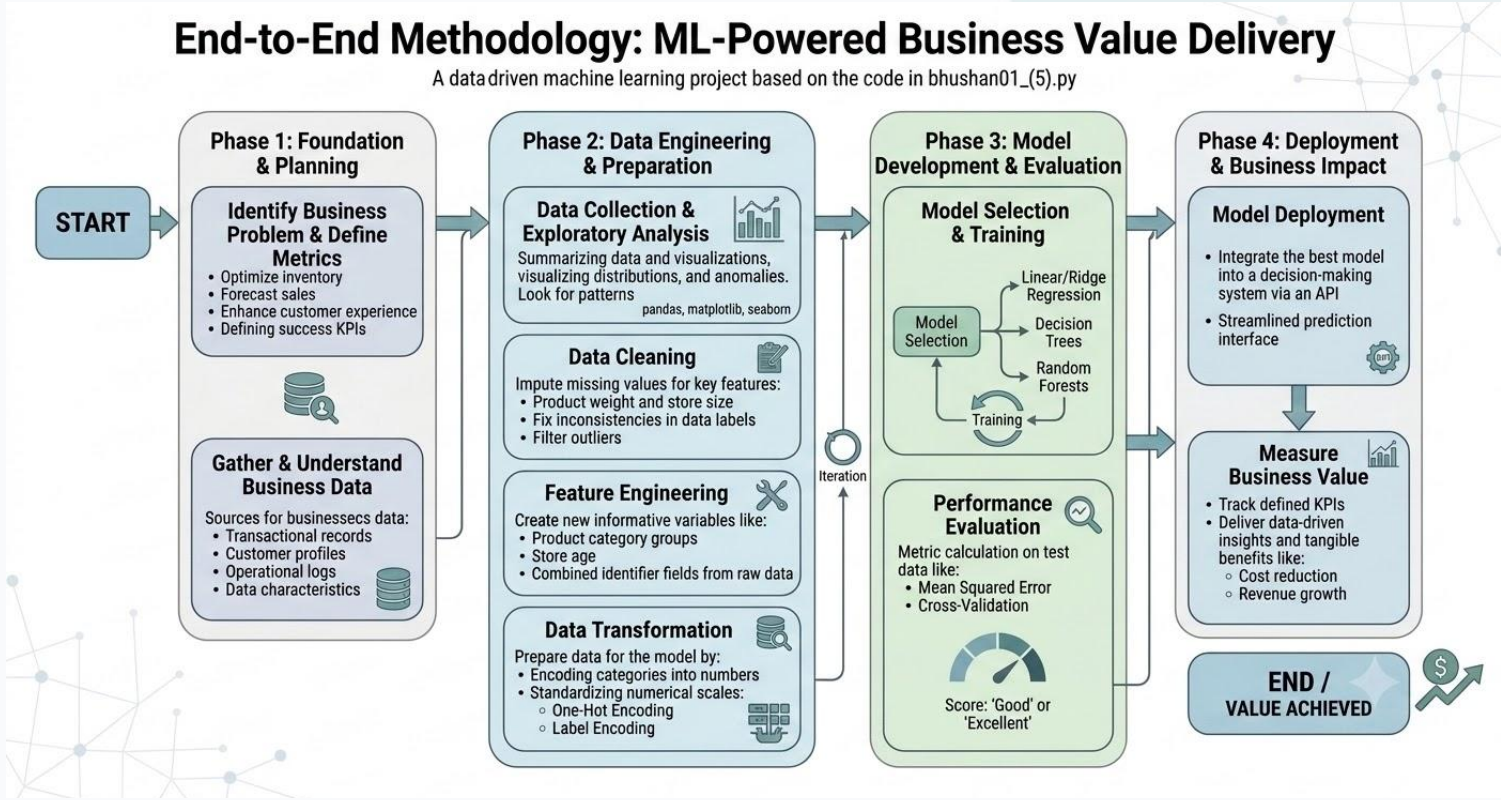
**90+ days?**

Statistical noise; premature intervention.

**31-60 days:**

Perfect window. CRM system can trigger targeted retention while persuadable.

image source : https://gemini.google.com/share/25085054928c




We know a $30 handset subsidy will keep customers. However, giving a $30 subsidy to *every* customer with an old phone is financially catastrophic.

It results in millions of dollars wasted on "False Positives"—customers who would have stayed anyway without the subsidy.

### The Strategic Objective

Proactively intercept high-value customers with an enticing **$30 upgrade subsidy**.

### The Financial Bottleneck

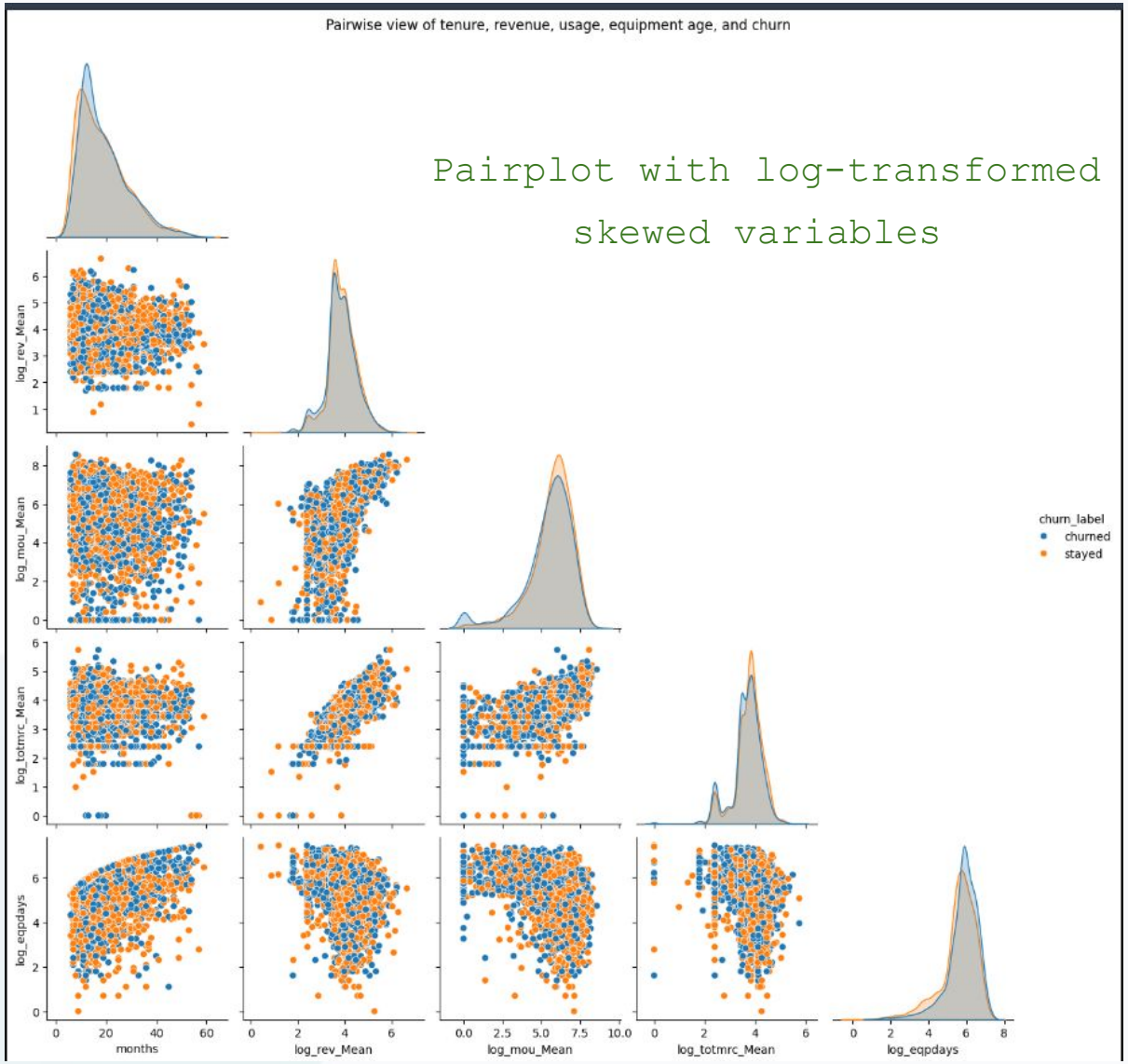
Blanket rollouts cause capital waste on **False Positives**.

### The Predictive Mandate

A mathematical model is required to target ***only*** high ROI investment.

Source: Bhushan01, Google Colab Notebook,Section 3: FEATURE ENGINEERING & AI MODEL TRAINING,Section 6: ADVANCED FEATURE ENGINEERING & IMPUTATION, Section 2: EXPLORATORY DATA ANALYSIS ( image source )

# FEATURE ENGINEERING




### The "Goldilocks Zone" Target

Model trained to predict churn specifically in the **31-60 day window**.

- ***Why?*** 0-30 days is too late (porting initiated). 90+ days is too noisy.
- Provides lead time for CRM to trigger targeted retention while the customer is persuadable.

### Advanced Feature Engineering

Distinct financial and technical metrics engineered:

`revenue_per_minute`: Network usage efficiency.

`total_failed_calls`: Technical friction evaluation.

### Data Integrity & Leakage Prevention

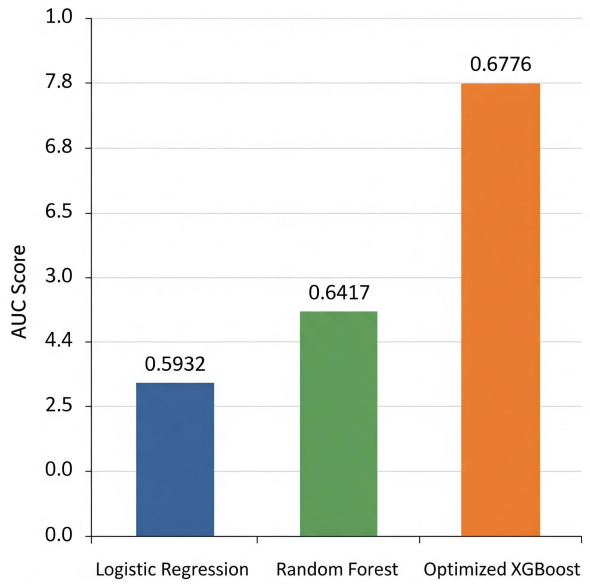
Imputation executed under strict, leak-free conditions.

Data split ***before*** processing to ensure simulation reflects real-world, unseen future performance.

Source: Bhushan01, Google Colab Notebook,Section 7: ALGORITHMIC COMPARISON & HYPERPARAMETER TUNING , Section 16: ROC Curve:

# Model Selection & Technical Validation

We aggressively tested multiple algorithms (Logistic Regression, Random Forest) before selecting our final production model.

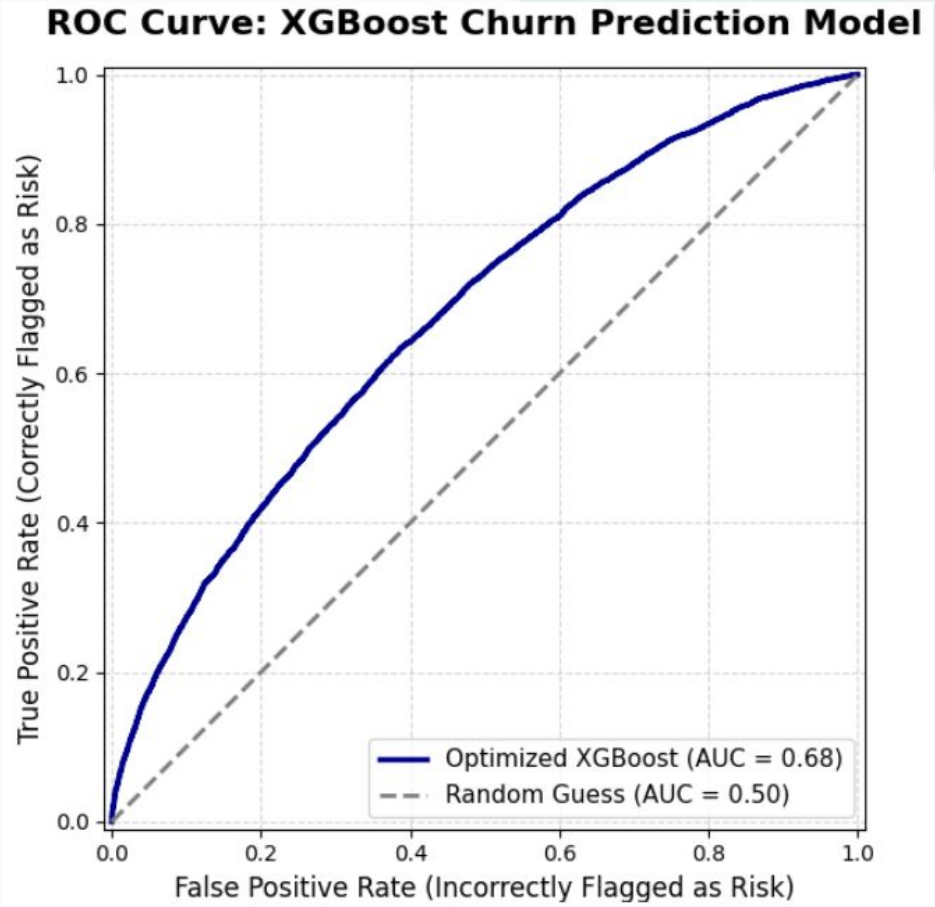



**Evaluation Metric: ROC-AUC**

**FOR Xboost : ROC-AUC = 0.6776**

Measures the model's ability to distinguish between a loyal customer and a churner.

**A robust, commercially viable score that significantly outperforms baseline guessing and legacy linear models.**

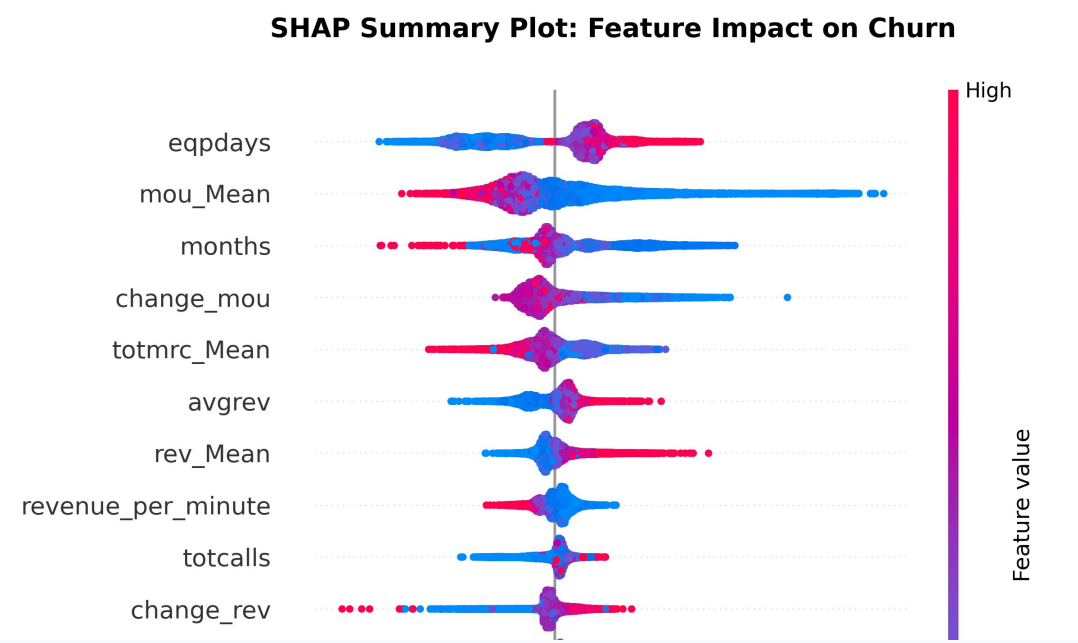


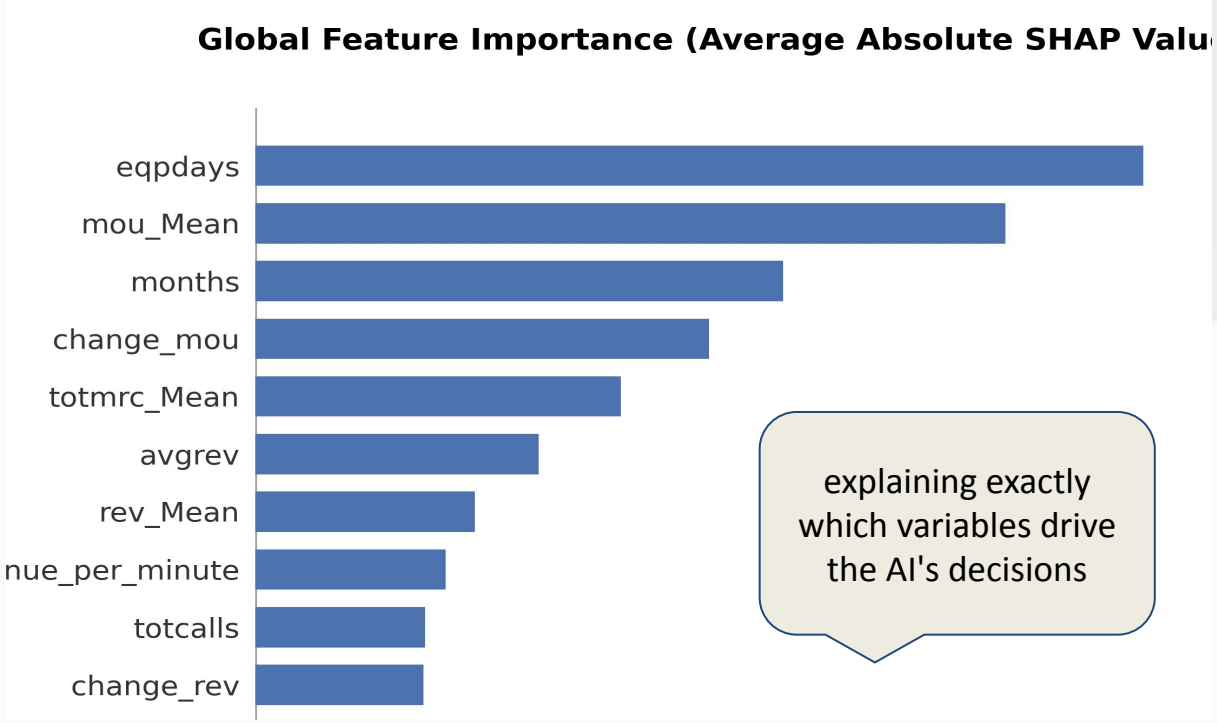

**The Selected Engine: Optimized XGBoost**

**Why XGBoost?** Traditional models (like Logistic Regression) draw straight lines through data. XGBoost excels at identifying complex, non-linear thresholds and interactions—such as the exact intersecting moment a customer hits the "12-Month Cliff" while exhibiting specific revenue behaviors.

Source:[1] Bhushan01, Google Colab Notebook,Section 14: SHAP Analysis , [2] https://chatgpt.com/share/6a33ff2f-7228-83ee-ae0d-808969e8d4af

# Model Interpretability (SHAP Analysis)




### The Key Risk Vectors

The SHAP summary confirms the top two most influential features for churn risk are **eqpdays** (equipment age) and **months** (tenure).

### Executive Alignment

AI successfully aligns with our core business hypothesis, making decisions based on logical, high-impact revenue and hardware lifecycle vectors.

### Demanding Transparency

Corporate governance requires explainable decisions. We used SHAP analysis to measure the exact global impact of each feature.

# Business Proposal – THE CRRI FRAMEWORK

NOTE : CRRI FOR OUR DATA IS IN SECTION 15 OF NOTEBOOK

## Translating Machine Learning Probabilities into an Actionable Financial Index

**Introducing the Customer Risk & Revenue Index (CRRI):** Raw churn probabilities are not enough to guide a budget. We developed the CRRI framework to merge algorithmic risk with actual financial exposure.

**The Mathematical Logic:** The index calculates a rolling Expected Revenue Loss metric for every individual subscriber across a 6-month operational horizon.

```
Expected Revenue Loss = XGBoost Churn Probability * (Subscriber ARPU * 6 Months)
```

**Portfolio Decile Segmentation:**

The entire customer base is automatically sorted into 10 equal tiers (Decile 1 = Safest, lowest financial risk; Decile 10 = Highest expected revenue loss). Marketing budgets are then surgically deployed *only* to the highest-exposure deciles, completely eliminating capital spend on low-value or low-risk accounts.

### What is the CRRI?

The Customer Risk & Revenue Index (CRRI) is a ranking system we created to make sure we don't waste a single dollar of our marketing budget. Instead of guessing who might leave, we use a formula to calculate the "Expected Revenue Loss" for every single customer.

**How the Ranking Works (1 to 10):**

- **Deciles 1–7 (The "Safe" Zone):** These customers are either unlikely to leave or their bill is too small to justify a $30 discount. We do nothing here.
- **Deciles 8–10 (The "Danger" Zone):** These are our high-value customers who the AI predicts are about to leave because of their old phones. This is where we spend our money.

**The Strategy: Surgical Targeting**

- **Stop the Waste:** In the past, we might have sent offers to everyone.
- **Focus on Impact:** We target **only the top 30%** of at-risk customers.
- **The Bottom Line:** We turn a high-cost campaign into a **$2.8M profit engine.**

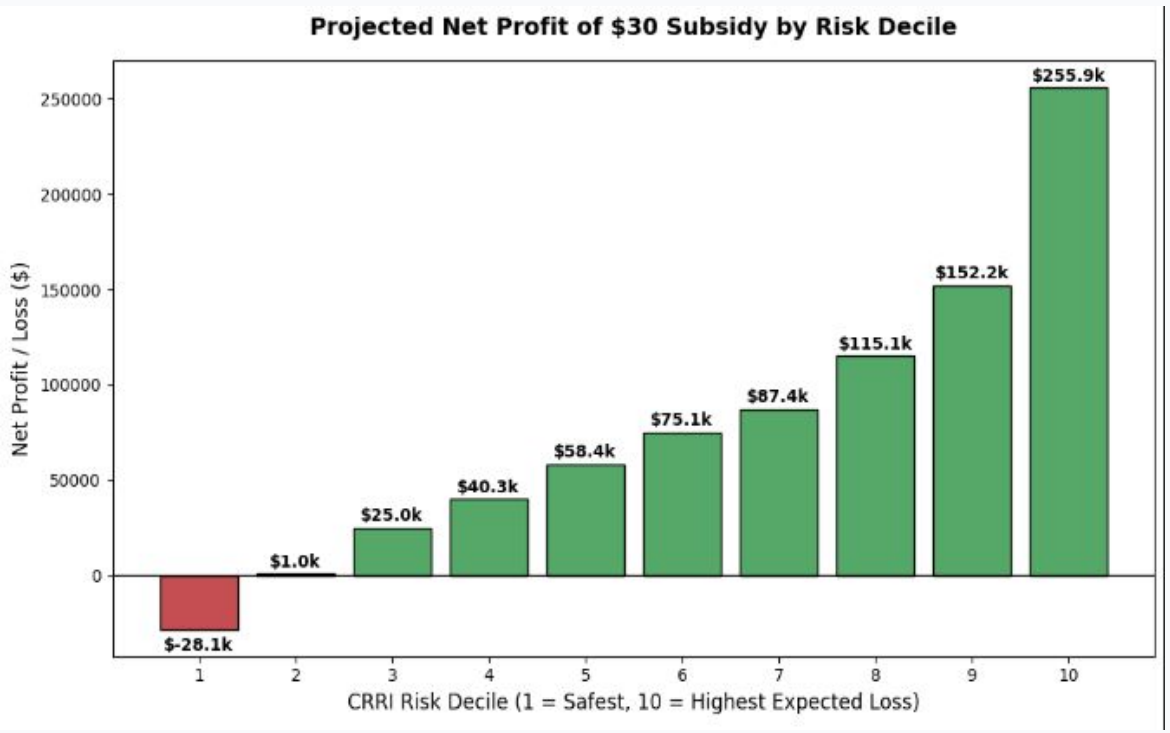
Generating Customer Risk & Revenue Index (CRRI)...

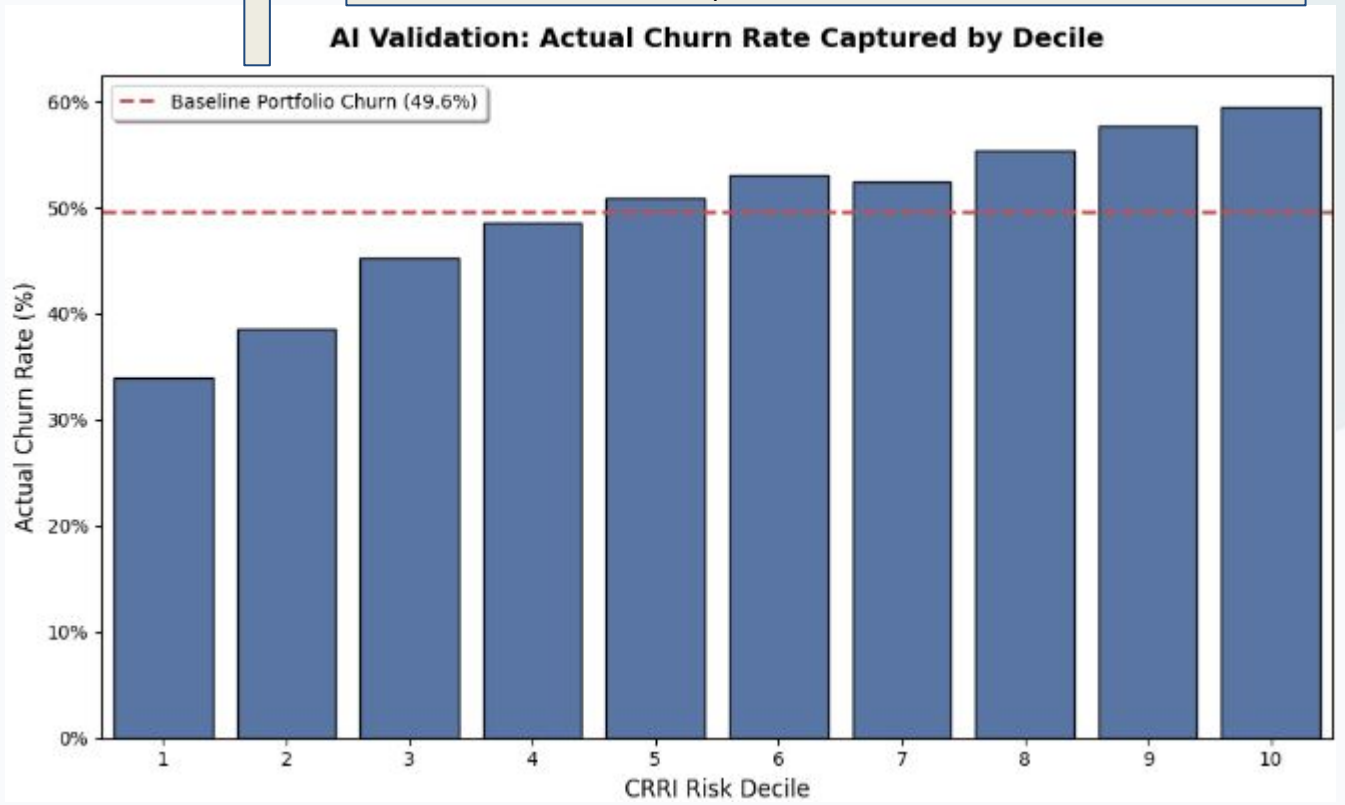
| | CRRI_Decile | targeted_customers | avg_churn_prob | actual_churn_rate | avg_success_rate | avg_monthly_revenue | campaign_cost | retained_customers | revenue_saved | net_profit |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2000 | 0.36 | 0.34 | 0.32 | 24.44 | 60000.0 | 217.42 | 31883.37 | -28116.63 |
| 1 | 2 | 2000 | 0.40 | 0.39 | 0.38 | 34.82 | 60000.0 | 291.88 | 60978.45 | 978.45 |
| 2 | 3 | 2000 | 0.45 | 0.45 | 0.42 | 37.71 | 60000.0 | 375.76 | 85013.28 | 25013.28 |
| 3 | 4 | 2000 | 0.49 | 0.49 | 0.41 | 41.69 | 60000.0 | 400.88 | 100284.29 | 40284.29 |
| 4 | 5 | 2000 | 0.50 | 0.51 | 0.40 | 48.18 | 60000.0 | 409.64 | 118419.37 | 58419.37 |
| 5 | 6 | 2000 | 0.53 | 0.53 | 0.40 | 52.75 | 60000.0 | 426.69 | 135054.44 | 75054.44 |
| 6 | 7 | 2000 | 0.53 | 0.52 | 0.39 | 60.48 | 60000.0 | 406.20 | 147398.38 | 87398.38 |
| 7 | 8 | 2000 | 0.55 | 0.55 | 0.38 | 68.84 | 60000.0 | 423.86 | 175063.93 | 115063.93 |
| 8 | 9 | 2000 | 0.57 | 0.58 | 0.38 | 81.72 | 60000.0 | 432.75 | 212190.48 | 152190.48 |
| 9 | 10 | 2000 | 0.60 | 0.60 | 0.35 | 125.78 | 60000.0 | 418.64 | 315931.31 | 255931.31 |

Source: Bhushan01, Google Colab Notebook,Section 15: ADVANCED ANALYTICS & CRRI FRAMEWORK
IMAGE: Bhushan01, Google Colab Notebook,Section 15: ADVANCED ANALYTICS & CRRI FRAMEWORK

# Verifying the Proposal: Lift Chart Validation

THE MODEL VALIDATION (Churn Lift Chart)" which proves that the users the model flags in Decile 10 actually churn at a massive 60% rate compared to the baseline.




- **Empirical Lift Validation**: To verify the CRRI framework before deploying capital, we mapped it directly against our unseen holdout validation data. The results are definitive: the actual, real-world churn rate spikes dramatically and exclusively in the top deciles.
- **Concentrating the Risk:** While the average baseline portfolio churn sits at a manageable level, Decile 10 captures a massive real-world churn rate, and Decile 9 follows closely behind.
- **The Structural Proof:** This empirical lift chart proves that the machine learning model successfully concentrates our defection risk into a small, highly targeted pool. By deploying campaigns to these top deciles, we hit the exact users causing our revenue leakage.

Source: Bhushan01, Google Colab Notebook,Section 15: ADVANCED ANALYTICS & CRRI FRAMEWORK

# THE $30 INTERVENTION STRATEGY

## Surgically Deploying the $30 Subsidy Against Capital Waste

**Standard 0.50 Threshold Profit: $2,757,087.36**
**OPTIMAL Probability Threshold: 0.30**
**MAXIMIZED Enterprise Profit: $3,552,728.16**

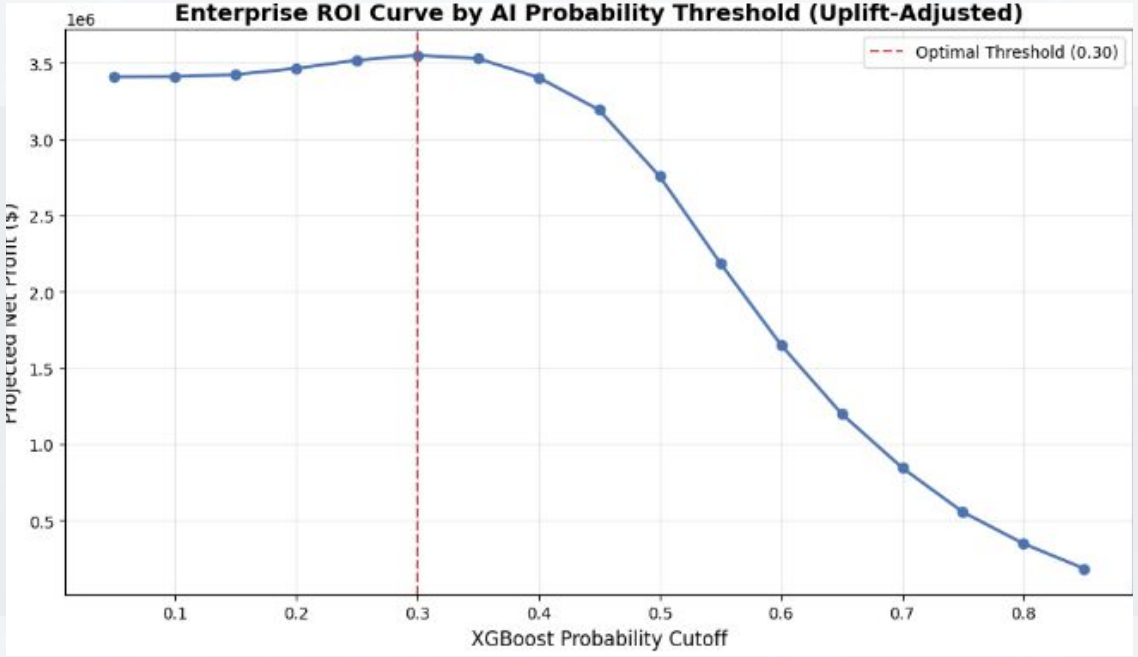
### Campaign Parameters

Targeted, automated retention offering a $30 handset upgrade subsidy to subscribers in top CRRI deciles.

### Capital Preservation

Ensures capital deployment only where upgrade probability aligns with high revenue-at-risk.

Section 13: Unique ROI Dashboard Visual: The interactive HTML dashboard output, demonstrating how adjusting the $30 subsidy or success rates dynamically impacts the bottom line.




Source:[images] Bhushan01, Google Colab Notebook,Section 8: THE CRMS FINANCIAL SIMULATION

### The Action Plan

**The Offer: We proactively offer a $30 device upgrade subsidy to our highest-risk, highest-value customers to lock them into a new contract before they port out.**

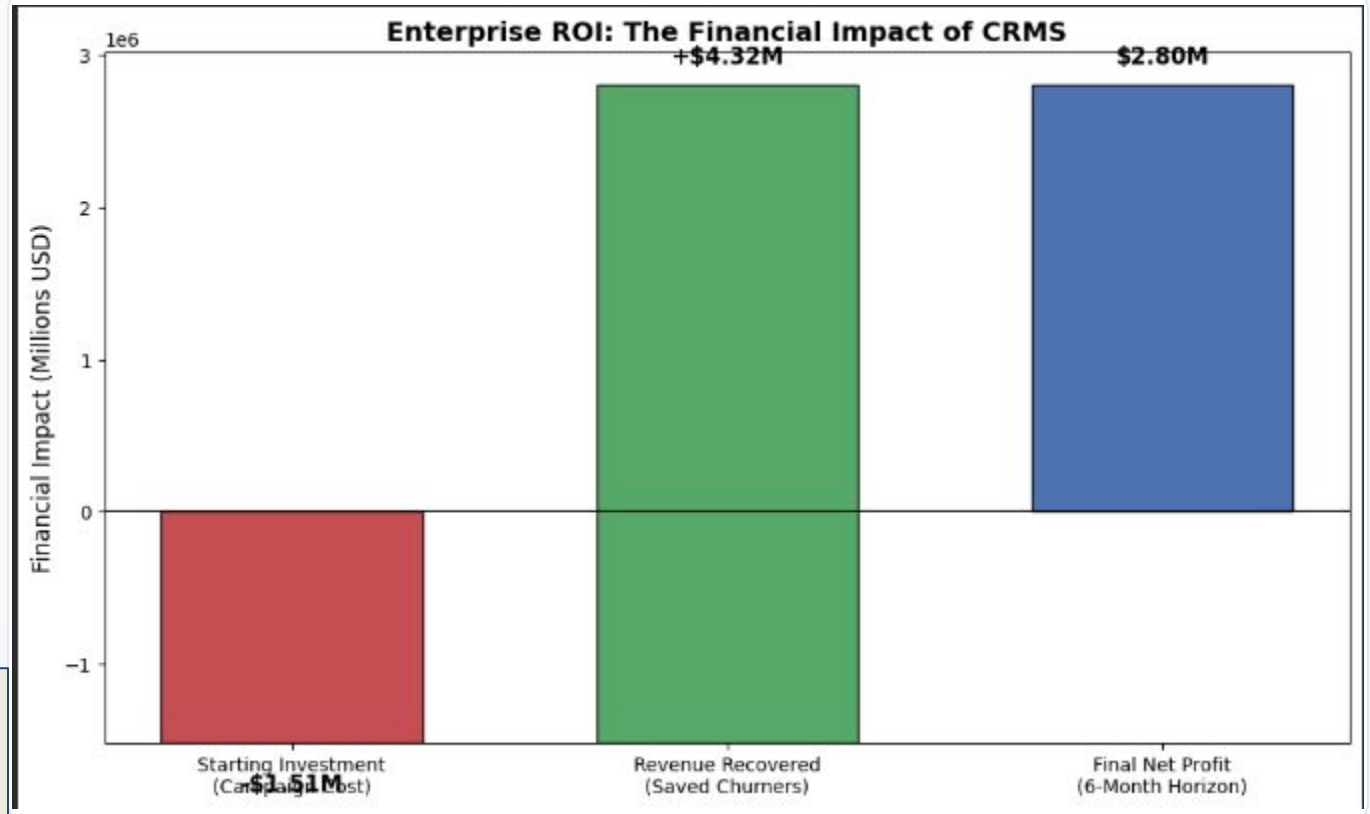
**The Reality Check (Conversion Math): We do not naively assume 100% of people will say yes. Our financial model only spends budget where it will actually convert:**

- **High Acceptance (60%): Customers with devices older than 1 year. Their batteries are dying, they are frustrated, and they are highly likely to take the $30 deal.**
- **Low Acceptance (20%): Customers with newer devices. They do not care about an upgrade yet. We ignore them to protect our margins.**

**The Bottom Line: We stop wasting marketing dollars on people who don't want a new phone, and surgically target the $30 offer to the exact people desperate for one.**

# Financial ROI & Enterprise Impact

## Surgically Maximizing Enterprise Profitability at the Optimal Probability Threshold




Source [image]: Bhushan01, Google Colab Notebook,Section 9: EXECUTIVE FINANCIAL WATERFALL CHART

**The Failure of Un-Targeted Campaigns**

If the business blindly deploys a $30 subsidy to all users, the mass marketing costs and massive False Positives result in severe net losses, completely draining company resources.

**The AI Optimization Victory**

By utilizing our Optimized XGBoost engine to restrict targeting exclusively to the high-yield CRRI Deciles, the campaign dynamics radically invert.

### Scaled Enterprise ROI (100k Users)

Campaign Cost: $302,820.00 targeted spend

Revenue Recovered: $863,781.58 saved

**Net Profit Lift: $2,804,807.88**

EXECUTIVE FINANCIAL WATERFALL CHART: The visual bridge showing Starting Investment -> Revenue Recovered -> Final Net Profit.

Source: Bhushan01, Google Colab Notebook,Section 4: ALGORITHMIC VALIDATION & FINANCIAL SIMULATION

# THE IMPLEMENTATION ROADMAP

## A Phased Integration Strategy to Secure $2.8M in Retained Value

**PHASE 1**

**Silent Pipeline Integration (Days 1–30):**

Embed the validated XGBoost data pipeline natively into our current data infrastructure. Run the scoring model silently in the background to automatically tag incoming customer records with their respective CRRI deciles without initiating active marketing outreach.

**PHASE 2**

**Live A/B Validation Campaign (Days 31–60):**

Launch a live field test targeting a controlled subset of users in CRRI Deciles 8-10. Maintain a strict, non-contacted control group to empirically verify the modeled 60% upgrade acceptance rate and measure true revenue lift.

**PHASE 3**

**Automated CRM Scale-Up (Days 61+):**

Upon successful field validation, fully automate the campaign. Program the CRM database to automatically trigger the $30 handset subsidy offer the moment a high-value account crosses into the high-risk CRRI tiers.

**Risk Mitigation & Model Governance:**

Technical environments suffer from *concept drift*—customer behaviors and device trends shift over time. To protect our profit margins against accuracy degradation, we will implement an automated monthly model retraining loop using freshly logged portfolio data.